

# UTS Communication Protocol

Case 2: Product/Inventory API Analysis

ADITYA WARDANA (25120500001)

Sains Data Profesional - Universitas

Cakrawala

---

# Tujuan & Skenario Bisnis

## 🎯 Tujuan

Verifikasi komunikasi client-server melalui HTTP protocol, observasi headers/status code, dan analisis JSON data encoding pada sistem inventaris.

## 👛 Skenario

Simulasi manajemen stok barang: Melakukan penarikan data (GET) serta penambahan produk baru (POST) pada database backend melalui API.

# Endpoint & Design Request

 **GET All:** /api/products - Menampilkan seluruh daftar inventaris.

 **GET Detail:** /api/products/1 - Menampilkan spesifik produk ID 1.

 **POST Data:** /api/products - Injeksi payload JSON (Smart Sensor & NodeMCU).

 **Headers:** Menggunakan Content-Type: application/json untuk validasi POST.

---

# Evidence GET & POST

## Eksekusi Sukses

Seluruh request dieksekusi via PowerShell menggunakan curl.exe.

GET: Menghasilkan HTTP 200 OK.

POST: Menghasilkan HTTP 201 Created.

Data dikembalikan dalam format terstruktur (JSON).

```
Windows x Window: x Settings x + v - □ x
PS C:\Users\warda> curl.exe -i -X GET "http://localhost:8088/api/products"
HTTP/1.0 200 OK
Server: CommProtocolsDemo/1.0 Python/3.13.3
Date: Sun, 07 Jun 2026 12:36:07 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 642
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type

{
  "resource": "products",
  "case": "Product",
  "count": 3,
  "methods": [
    "GET",
    "POST",
    "PUT",
    "PATCH",
    "DELETE"
  ],
  "canonicalPath": "/api/products",
  "itemPathExample": "/api/products/1",
  "data": [
    {
      "id": 1,
      "name": "Laptop Pro 14",
      "category": "computer",
      "price": 14500000,
      "stock": 5
    },
    {
      "id": 2,
      "name": "Temperature Sensor DHT22",
      "category": "iot",
      "price": 85000,
      "stock": 30
    },
    {
      "id": 3,
      "name": "Wireless Router AC1200",
      "category": "networking",
      "price": 475000,
      "stock": 10
    }
  ]
}
PS C:\Users\warda>
```

# Evidence GET & POST

```
PS C:\Users\warda> curl.exe -i -X POST "http://localhost:8088/api/products" -H "Content-Type: application/json" -d '{"name": "Smart Sensor", "category": "iot", "price": 175000, "stock": 12}'
HTTP/1.0 201 Created
Server: CommProtocolsDemo/1.0 Python/3.13.3
Date: Sun, 07 Jun 2026 12:48:04 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 137
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
Location: /api/products/4

{
  "id": 4,
  "name": "Smart Sensor",
  "category": "iot",
  "price": 175000,
  "stock": 12,
  "createdAt": "2026-06-07T19:48:04+0700"
}
PS C:\Users\warda> |
```

```
PS C:\Users\warda> curl.exe -i -X GET "http://localhost:8088/api/products/1"
HTTP/1.0 200 OK
Server: CommProtocolsDemo/1.0 Python/3.13.3
Date: Sun, 07 Jun 2026 12:39:13 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 101
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type

{
  "id": 1,
  "name": "Laptop Pro 14",
  "category": "computer",
  "price": 14500000,
  "stock": 5
}
PS C:\Users\warda> |
```

```
PS C:\Users\warda> curl.exe -i -X POST "http://localhost:8088/api/products" -H "Content-Type: application/json" -d '{"name": "Microcontroller NodeMCU", "category": "iot", "price": 65000, "stock": 25}'
HTTP/1.0 201 Created
Server: CommProtocolsDemo/1.0 Python/3.13.3
Date: Sun, 07 Jun 2026 12:49:02 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 147
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
Location: /api/products/5

{
  "id": 5,
  "name": "Microcontroller NodeMCU",
  "category": "iot",
  "price": 65000,
  "stock": 25,
  "createdAt": "2026-06-07T19:49:02+0700"
}
PS C:\Users\warda> |
```

# Kendala & Solusi

```
Windows X Window: X Settings X + - □ X
Sensor\, \category\: \iot\, \price\: 175000, \stock\: 12}
PS C:\Users\warda> curl.exe -i -X POST "http://localhost:8088/api/products
" -H "Content-Type: application/json" -d '{"name": "Smart Sensor", "catego
ry": "iot", "price": 175000, "stock": 12}'
HTTP/1.0 400 Bad Request
Server: CommProtocolsDemo/1.0 Python/3.13.3
Date: Sun, 07 Jun 2026 12:45:07 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 59
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type

{
  "error": "Invalid JSON body",
  "raw": "{name: Smart
}curl: (3) unmatched close brace/bracket in URL position 48:
Sensor, category: iot, price: 175000, stock: 12}
PS C:\Users\warda> |
```

## Solusi: Escape String

Menggunakan karakter pelolosan \" pada setiap key dan value JSON untuk menjaga integritas payload saat transmisi.

```
-d '{"name\: \"Smart Sensor\"...}'
```

# Analisis Response

| Komponen    | Hasil Analisis Teknis                            |
|-------------|--------------------------------------------------|
| Status Code | 200 OK (GET) & 201 Created (POST Berhasil).      |
| Encoding    | JSON terencode UTF-8 (Header: application/json). |
| Data Types  | String (name, cat) & Integer (id, price, stock). |
| CORS        | Access-Control-Allow-Origin: * (Izin publik).    |

# Evidence Traffic Capture Wireshark

\*Adapter for loopback traffic capture

Berkas Sunting Lihat Go Tangkap Analisis Statistik Telefoni Nirkabel Perangkat Bantuan

tcp.port == 8088 and http

| No. | Time         | Source    | Destination | Protocol | Length | Info                                                 |
|-----|--------------|-----------|-------------|----------|--------|------------------------------------------------------|
| 6   | 0.206478200  | 127.0.0.1 | 127.0.0.1   | HTTP     | 134    | GET /api/products HTTP/1.1                           |
| 10  | 0.208146200  | 127.0.0.1 | 127.0.0.1   | HTTP/1.0 | 686    | HTTP/1.0 200 OK, JSON (application/json)             |
| 21  | 18.067987200 | 127.0.0.1 | 127.0.0.1   | HTTP     | 136    | GET /api/products/1 HTTP/1.1                         |
| 25  | 18.069638600 | 127.0.0.1 | 127.0.0.1   | HTTP/1.0 | 145    | HTTP/1.0 200 OK, JSON (application/json)             |
| 36  | 30.350503000 | 127.0.0.1 | 127.0.0.1   | HTTP/1.0 | 260    | POST /api/products HTTP/1.1, JSON (application/json) |
| 40  | 30.353458400 | 127.0.0.1 | 127.0.0.1   | HTTP/1.0 | 181    | HTTP/1.0 201 Created, JSON (application/json)        |
| 51  | 41.049319700 | 127.0.0.1 | 127.0.0.1   | HTTP/1.0 | 270    | POST /api/products HTTP/1.1, JSON (application/json) |
| 55  | 41.052281400 | 127.0.0.1 | 127.0.0.1   | HTTP/1.0 | 191    | HTTP/1.0 201 Created, JSON (application/json)        |

> POST /api/products HTTP/1.1\r\nHost: localhost:8088\r\nUser-Agent: curl/8.19.0\r\nAccept: \*/\*\r\nContent-Type: application/json\r\nContent-Length: 83\r\n\r\n[Response in frame: 55]  
[Full request URI: http://localhost:8088/api/products]  
File Data: 83 bytes

JavaScript Object Notation: application/json

Object

- Member: name
  - [Path with value: /name:Microcontroller NodeMCU]
  - [Member with value: name:Microcontroller NodeMCU]
  - String value: Microcontroller NodeMCU
  - Key: name
  - [Path: /name]
- Member: category
  - [Path with value: /category:iot]
  - [Member with value: category:iot]
  - String value: iot
  - Key: category
  - [Path: /category]
- Member: price
  - [Path with value: /price:65000]
  - [Member with value: price:65000]
  - Number value: 65000
  - Key: price
  - [Path: /price]
- Member: stock
  - [Path with value: /stock:25]
  - [Member with value: stock:25]
  - Number value: 25
  - Key: stock

0000 02 00 00 00 45 00 01 0a 58 38 40 00 80 06 a3 b3 .....E...X8@.....  
0010 7f 00 00 01 7f 00 00 01 f6 6f 1f 98 1a 4f 2b 47 .....o...O+G  
0020 a4 bf 6c 46 50 18 00 ff 3e 71 00 00 50 4f 53 54 ...lFP...>q..POST  
0030 20 2f 61 70 69 2f 70 72 6f 64 75 63 74 73 20 48 /api/pr oducts H  
0040 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6c TTP/1.1 ..Host: l  
0050 6f 63 61 6c 68 6f 73 74 3a 38 30 38 38 0d 0a 55 ocalhost :8088..U  
0060 73 65 72 2d 41 67 65 6e 74 3a 20 63 75 72 6c 2f ser-Agen t: curl/  
0070 38 2e 31 39 2e 30 0d 0a 41 63 63 65 70 74 3a 20 8.19.0.. Accept:  
0080 2a 2f 2a 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 \*/\*. Con tent-Typ  
0090 65 3a 20 61 70 70 6c 69 63 61 74 69 6f 6e 2f 6a e: appli cation/j  
00a0 73 6f 6e 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e son..Con tent-Len  
00b0 67 74 68 3a 20 38 33 0d 0a 0d 0a 7b 22 6e 61 6d gth: 83- ...{"nam  
00c0 65 22 3a 20 22 4d 69 63 72 6f 63 6f 6e 74 72 6f e": "Mic rocontro  
00d0 6c 6c 65 72 20 4e 6f 64 65 4d 43 55 22 2c 20 22 ller Nod eMCU", "  
00e0 63 61 74 65 67 6f 72 79 22 3a 20 22 69 6f 74 22 category ": "iot"  
00f0 2c 20 22 70 72 69 63 65 22 3a 20 36 35 30 30 30 , "price ": 65000  
0100 2c 20 22 73 74 6f 63 6b 22 3a 20 32 35 7d , "stock ": 25}

Hypertext Transfer Protocol: Protocol

Paket: 60 · Ditampilkan: 8 (13.3%) · Dijatuhkan: 0 (0.0%)

Profil: Default



# Evidence Traffic Capture Wireshark

\*Adapter for loopback traffic capture

Berkas Sunting Lihat Go Tangkap Analisis Statistik Telefoni Nirkabel Perangkat Bantuan

tcp.port == 8088 and http

| No. | Time         | Source    | Destination | Protocol  | Length | Info                                                 |
|-----|--------------|-----------|-------------|-----------|--------|------------------------------------------------------|
| 6   | 0.206478200  | 127.0.0.1 | 127.0.0.1   | HTTP      | 134    | GET /api/products HTTP/1.1                           |
| 10  | 0.208146200  | 127.0.0.1 | 127.0.0.1   | HTTP/JSON | 686    | HTTP/1.0 200 OK, JSON (application/json)             |
| 21  | 18.067987200 | 127.0.0.1 | 127.0.0.1   | HTTP      | 136    | GET /api/products/1 HTTP/1.1                         |
| 25  | 18.069638600 | 127.0.0.1 | 127.0.0.1   | HTTP/JSON | 145    | HTTP/1.0 200 OK, JSON (application/json)             |
| 36  | 30.350503000 | 127.0.0.1 | 127.0.0.1   | HTTP/JSON | 260    | POST /api/products HTTP/1.1, JSON (application/json) |
| 40  | 30.353458400 | 127.0.0.1 | 127.0.0.1   | HTTP/JSON | 181    | HTTP/1.0 201 Created, JSON (application/json)        |
| 51  | 41.049319700 | 127.0.0.1 | 127.0.0.1   | HTTP/JSON | 270    | POST /api/products HTTP/1.1, JSON (application/json) |
| 55  | 41.052281400 | 127.0.0.1 | 127.0.0.1   | HTTP/JSON | 191    | HTTP/1.0 201 Created, JSON (application/json)        |

▼ Hypertext Transfer Protocol

- > POST /api/products HTTP/1.1\r\nHost: localhost:8088\r\nUser-Agent: curl/8.19.0\r\nAccept: \*/\*\r\nContent-Type: application/json\r\nContent-Length: 73\r\n\r\n[Response in frame: 40]  
[Full request URI: http://localhost:8088/api/products]  
File Data: 73 bytes

▼ JavaScript Object Notation: application/json

▼ Object

- ▼ Member: name
  - [Path with value: /name:Smart Sensor]
  - [Member with value: name:Smart Sensor]
  - String value: Smart Sensor
  - Key: name
  - [Path: /name]
- ▼ Member: category
  - [Path with value: /category:iot]
  - [Member with value: category:iot]
  - String value: iot
  - Key: category
  - [Path: /category]
- ▼ Member: price
  - [Path with value: /price:175000]
  - [Member with value: price:175000]
  - Number value: 175000
  - Key: price
  - [Path: /price]
- ▼ Member: stock
  - [Path with value: /stock:12]
  - [Member with value: stock:12]
  - Number value: 12

```
0000 02 00 00 00 45 00 01 00 58 2b 40 00 80 06 a3 ca .....E...X+@.....
0010 7f 00 00 01 7f 00 00 01 f5 bc 1f 98 6a 09 d2 06 .....j....
0020 65 36 dd 6f 50 18 00 ff bd 6c 00 00 50 4f 53 54 e6 oP...l..POST
0030 20 2f 61 70 69 2f 70 72 6f 64 75 63 74 73 20 48 /api/pr oducts H
0040 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6c TTP/1.1 -Host: l
0050 6f 63 61 6c 68 6f 73 74 3a 38 30 38 38 0d 0a 55 ocalhost :8088 -U
0060 73 65 72 2d 41 67 65 6e 74 3a 20 63 75 72 6c 2f ser-Agen t: curl/
0070 38 2e 31 39 2e 30 0d 0a 41 63 63 65 70 74 3a 20 8.19.0.. Accept:
0080 2a 2f 2a 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 */*. Con tent-Typ
0090 65 3a 20 61 70 70 6c 69 63 61 74 69 6f 6e 2f 6a e: appli cation/j
00a0 73 6f 6e 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e son..Con tent-Len
00b0 67 74 68 3a 20 37 33 0d 0a 0d 0a 7b 22 6e 61 6d gth: 73- ...{"nam
00c0 65 22 3a 20 22 53 6d 61 72 74 20 53 65 6e 73 6f e": "Sma rt Senso
00d0 72 22 2c 20 22 63 61 74 65 67 6f 72 79 22 3a 20 r", "cat egory":
00e0 22 69 6f 74 22 2c 20 22 70 72 69 63 65 22 3a 20 "iot", " price":
00f0 31 37 35 30 30 30 2c 20 22 73 74 6f 63 6b 22 3a 175000, "stock":
0100 20 31 32 7d 12}
```

Hypertext Transfer Protocol: Protocol

Paket: 60 · Ditampilkan: 8 (13.3%) · Dijatuhkan: 0 (0.0%)

Profil: Default

# Analisis TRAFFIC CAPTURE (WIRESHARK)

| Komponen                 | Hasil Analisis Teknis                                                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Loopback Capture         | Pemantauan <i>traffic</i> komunikasi lokal (127.0.0.1) pada <i>port</i> 8088.                                                                                |
| Request-Response Sinkron | Paket HTTP berjalan lancar di atas protokol TCP tanpa deteksi <i>packet loss</i> atau anomali jaringan.                                                      |
| Inspeksi JSON Dissector  | Wireshark berhasil membedah objek <i>request</i> POST. <i>Payload</i> data "Smart Sensor" dan "Microcontroller NodeMCU" terbukti terserialisasi secara utuh. |
| Validasi Resolusi        | Membuktikan secara konkret bahwa implementasi <i>escape string</i> (\") berhasil menjaga integritas format JSON saat transmisi.                              |

# TERIMA KASIH

Kesimpulan: Komunikasi client-server via REST API sukses diimplementasikan dan dianalisis.

<https://github.com/AdityaaWardanaa/communication-protocol-uts>

---